

SANs and VIP / Load Balancer for the Kubernetes API Server

Why they matter and why they belong together

Overview

SANs and a VIP or load balancer solve two different problems, and they work as a pair. The VIP gives clients **one stable address** that keeps working even when individual API servers fail. The SANs make that address **trusted**, so TLS verification succeeds no matter which API server answers behind it.

Benefits of the VIP or load balancer

High availability. With three control-plane nodes behind a VIP, losing one node does not break kubectl, kubelets, controllers, or CI pipelines. The load balancer health-checks each API server on /readyz and stops sending traffic to failed ones. Without a VIP, every client is pinned to one node's IP, and that node becomes a single point of failure even if you run several control planes.

Zero-downtime maintenance. You can upgrade, reboot, or replace control-plane nodes one at a time while the VIP keeps serving traffic through the others.

Load distribution. Requests from kubelets, controllers, and users are spread across API servers instead of all landing on one. This matters in large clusters with many nodes and watches.

Decoupling clients from node addresses. Kubeconfigs, kubelet configs, and external tools all reference the VIP. Nodes can be renumbered or rebuilt without touching any of them.

Benefits of SANs in this setup

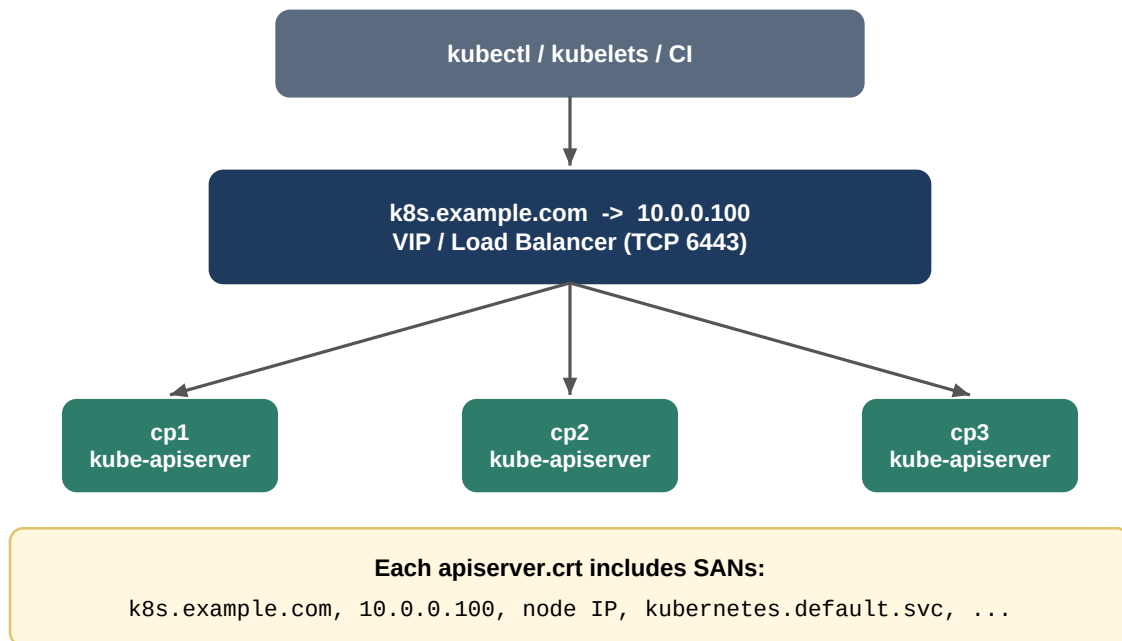
Certificates match the address clients use. Clients connect to the VIP or its DNS name, but the TLS session terminates on whichever API server is behind it. Each API server's certificate must list the VIP and DNS name as SANs, or every connection fails with an x509 error.

One certificate setup covers every access path. Internal VIP, public IP, DNS name, and VPN address can all be listed, so the same cluster is reachable securely from inside and outside.

No insecure workarounds. Correct SANs mean nobody needs insecure-skip-tls-verify, so full certificate verification protects against man-in-the-middle attacks.

Why they belong together

A VIP without matching SANs fails TLS on every request. SANs without a VIP give you trusted certificates but still leave clients pointed at a single node, so there is no failover. Together they give you an endpoint that is stable, highly available, and verified.



Example: kubeadm init for an HA control plane

```
sudo kubeadm init \
  --control-plane-endpoint "k8s.example.com:6443" \
  --apiserver-cert-extra-sans "k8s.example.com,10.0.0.100,203.0.113.5" \
  --upload-certs \
  --pod-network-cidr 10.244.0.0/16
```

Verify the SANs after init:

```
openssl x509 -in /etc/kubernetes/pki/apiserver.crt -noout -text \
  | grep -A1 "Subject Alternative Name"
```

Common ways to provide the VIP

On bare metal, the usual choices are **kube-vip** (runs as a static pod and announces the VIP via ARP or BGP) or **keepalived plus HAProxy** on the control-plane nodes or dedicated machines. In the cloud, you would typically use the provider's TCP load balancer, such as AWS NLB, Azure Load Balancer, or a GCP internal load balancer.

Practical recommendation

Use a DNS name as `--control-plane-endpoint`, point it at the VIP, and include both the name and the VIP IP as SANs. That way you can change the VIP itself later by updating DNS alone.